

idea in 1996, and after perfecting their technology and approach to repurposing plastic waste in asphalt-ing roads, they made propositions to various road builders but to no avail. After toiling for a while, they approached the Central Road Research Institute of India (CRRI) and Indian Road Congress (IRC), and finally, their innovation was cleared for use.

Finally, in 2002, the plastic-induced roads were constructed in Bengaluru as a trial. Now, the startup has constructed a total of 3,200 km of plastic-induced roads in Bengaluru alone which consumed 15,000 tonnes of plastic waste. Khan stated, "Using plastic in roads will increase their lifespan by three times and the cost will be driven down as well. If plastic is used to reconstruct every road in India, almost 1.5 crore tonnes of plastic waste can be redirected here".

To make this blend of plastic and tar coal, the startup designed their own machinery. However, the ones fabricated in 1996 are now obsolete, so they have created new automated machines with funding from companies such as Dr Reddy's Laboratories, FCL, ONGC and more.

The Prime Minister of India has allocated K.K. Plastic Waste Management a huge set up in Gurgaon. They're also in talks of setting up units at Noida, Hyderabad and Panvel in a few years which would allow them to take their initiative further.

Chakr Innovation

To deal with the menace of air pollution, Chakr Innovation, a Delhi-based startup, has developed an innovative technology called 'Chakr Shield'. This device is retrofitted on diesel generators which then capture 90 per cent of particulate emissions from said generators. These emissions, which would otherwise pollute the environment, are captured and converted into ink and paint, thus closing the carbon cycle. Shreya Kapoor, Marketing Manager at Chakr Innovations stated, "Our product deals with the problem of unrestrained emissions from diesel engine generators. Among these emissions, particulate matter (carbo-

naceous Nano-sized matter) pose an enormous risk to public health due to its carcinogenic nature".

The startup was founded by Kushagra Srivastava, Arpit Dhupar and Bharti Singhla, a team of IIT-Delhi graduates that came up with this novel technology back in their college days. Suffering from breathing problems to due air pollution right from his childhood, Dhupar's inspiration for this project is apparent. Dhupar, along with Srivastava and Singhla, thought of this idea when visiting a sugarcane stall. They noticed a wall had turned black from the emissions coming out of the diesel genset used by the vendor. This triggered their innovative instincts



Ink made from the residue of captured PM emissions from diesel generators by Chakr Innovations.

and they went ahead to develop a product that could capture such harmful emissions.

So far, Chakr Innovation has installed more than 80 Chakr Shields and they have captured a total of 500 kg of particulate matter. To break it down further, each device captures 17.5kg of particulate matter from a 25 kVA DG (diesel generator), running at 80 per cent load for 8 hours daily, over a period of one year. A single device can also purify around 255 billion litres of air in its lifetime. The more devices are installed, the lesser the PM levels in the air will become which will result in better visibility, air quality and reduced smog.

Suryagen Renewables

In collaboration with IISc researchers, Suryagen Renewables has devel-

oped an open-source solar water purifier that can convert water from any source, such as sea, river, ponds, rain, etc, into potable water. Impure water is collected in the contraption and then evaporated by solar energy and the vapours are condensed to pure water on a cold surface, eliminating bacteria, metal residue, arsenic, fluoride and other impurities.

Banyan Nation

Banyan Nation uses internally-developed thermal technology to clean and re-engineer 1,200 tonnes of plastic waste per annum. The plastic waste is restructured to become as close to virgin plastic as possible, thereby enabling reuse of the revamped plas-

tic, which they call 'Better Plastic', in manufacturing processes. Tata Motors and L'Oreal have both used Better Plastic in their auto components and product packaging respectively.

Ulta Chaata

Samit and Priya Choksi, a couple passionate about conservation and sustainability, created the Ulta Chaata. It is a patented canopy that captures rainwater and then filters it to provide potable water. The unit also works as a solar energy generation system and is IoT-enabled. A single unit can harvest 1,00,000 litres of water and capture energy with a maximum peak power of 1.5 kW.

If you want to be a 'green' hero yourself, start by taking small steps to reduce your own carbon footprint. Tell us about your efforts by writing to us. 